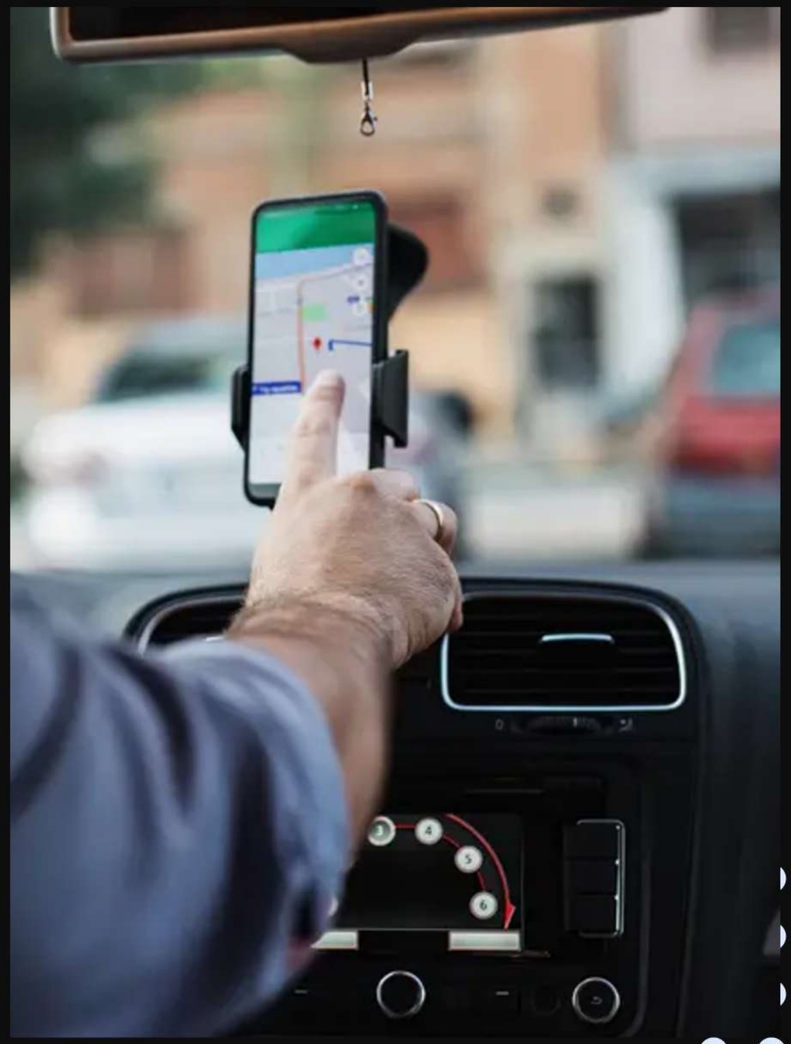
RIDE-HAILING ANALYTICS & FARE OPTIMIZATION

A Data-Driven Approach to Demand Patterns, Pricing Models, and Revenue Optimization

View Python Code

View Power BI Dashboard

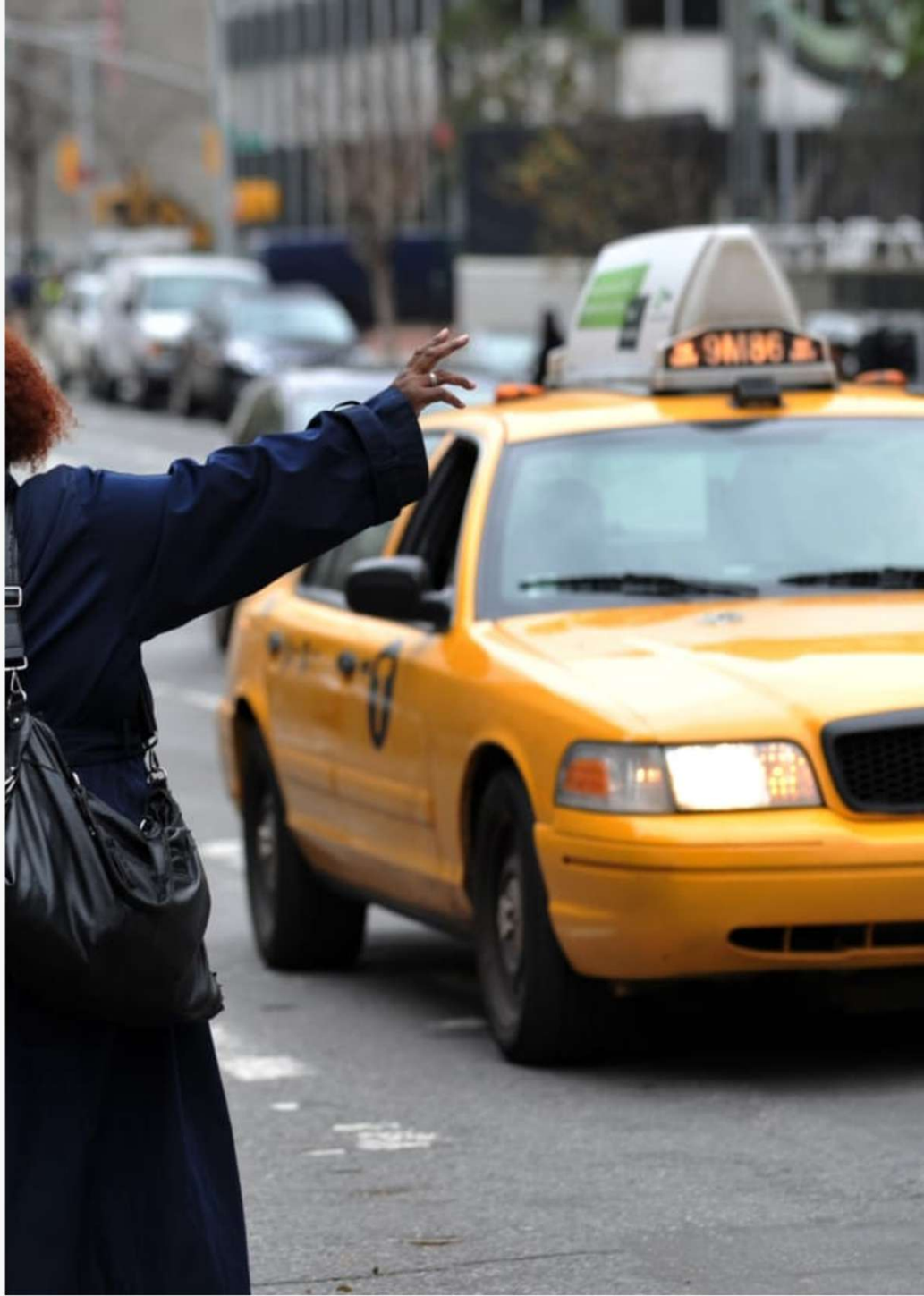
Nurhanim Insyirah binti Mohd Haniff



BUSINESS BACKGROUND

Ride-hailing is essential in NYC. With over 75% of Manhattan households living car-free, services like Uber act as a critical urban utility, providing 24/7 door-to-door connectivity in a city that "never sleeps."

While the raw dataset encompasses 200,000 trips from 2009–2015, this analysis intentionally focuses on the 2013–2015 window to capture a mature market environment. By identifying geospatial and temporal demand patterns, several strategies to maximize revenue and efficiency has been developed.



EXECUTIVE SUMMARY

01

Business Problem

- Uber revenue declined 56.03% from Q1 2013 to Q2 2015.
- The decline suggests inefficiencies in the existing pricing strategy.

02

Key Findings

- Distance is the most influential driver of fares
- Current pricing is not aligned with demand patterns
- Peak demand periods are underpriced
- Low demand periods are not sufficiently stimulated

03

Business Impact

- Optimized pricing is projected to increase revenue by 11.5% without changing base fares.

04

Recommendation

Implement demand-based pricing by:

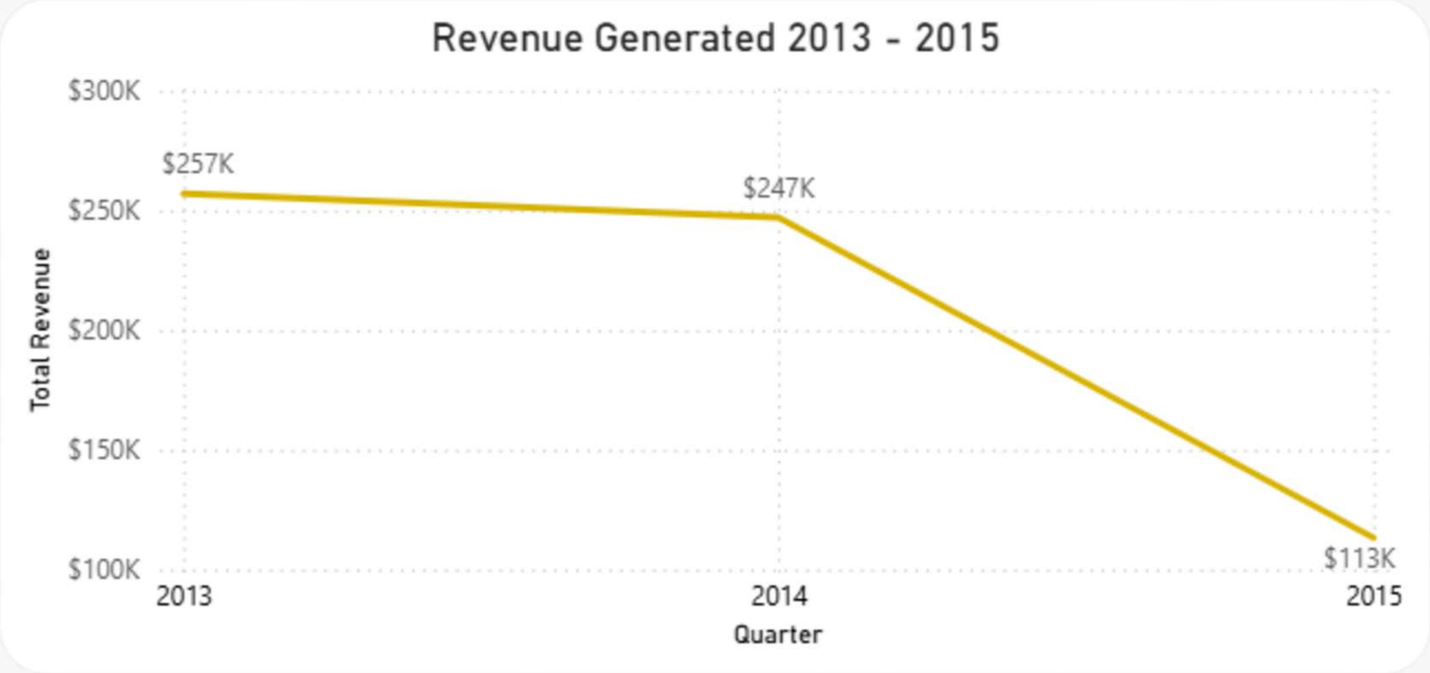
- Increasing fares during peak demand
- Applying discounts during low demand
- Adjusting pricing by time, location, and passenger count

REVENUE TREND

Total revenue **decreased by 56.03%** between 2013 to 2015.

The decline indicates that the existing pricing model may not be optimized for changing demand patterns.

To ensure our pricing model reflects a mature market environment, this project will focus on the 2013–2015 period. This timeframe represents Uber's most stabilized growth phase in NYC.



🎯 PROJECT OBJECTIVE

Thynk Unlimited combines years of experience with a deep understanding of the real estate market. Our personalized approach, attention to detail, and dedication to excellence make us the trusted choice for individuals and businesses alike.

Evaluate the existing pricing model and identify opportunities to **increase revenue by at least 10%** by the end of 2015.

KEY VARIABLES

Fare performance is influenced by several operational variables:

Trip distance

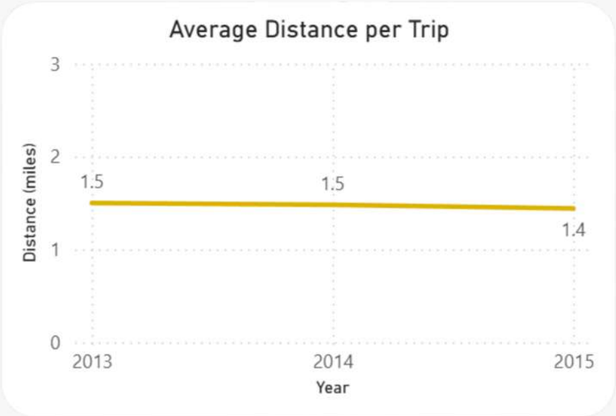
Passenger count

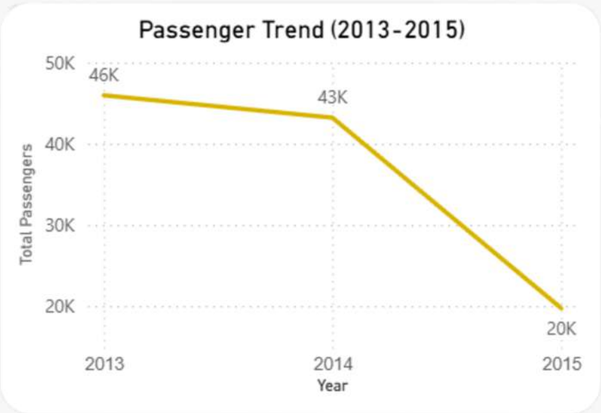
Number of orders

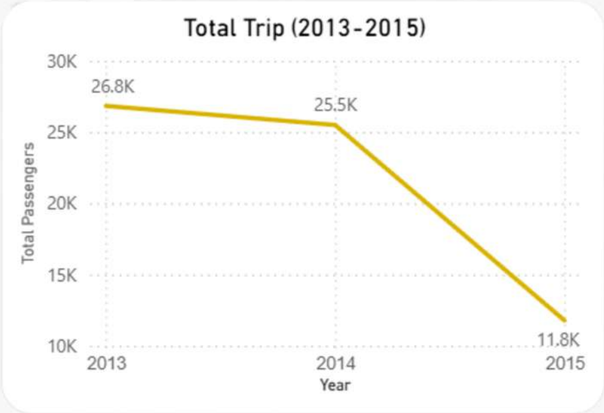
- Pickup hour
- Pickup location
- Pickup day

These variables were analyzed to evaluate the effectiveness of the existing pricing model.

OPERATIONAL TRENDS







Average distance per trip remained **stable**

Passenger count **decreased by 57%**

Total orders **decreased by 55%**

An analysis of the individual variables affecting current pricing is required, alongside an evaluation of the existing fare model (high, normal, and discounted) to identify opportunities for **revenue optimization.**

METHODOLOGY

Business Problem Identification

Background, problem statement, and objectives

Data Preparation

- Data cleaning using Python (missing values, duplicates, outliers)

Exploratory Data Analysis

Trend and pattern identification

Pricing Analysis

Linear regression and pricing evaluation

Impact Simulation

Revenue comparison between existing and optimized pricing

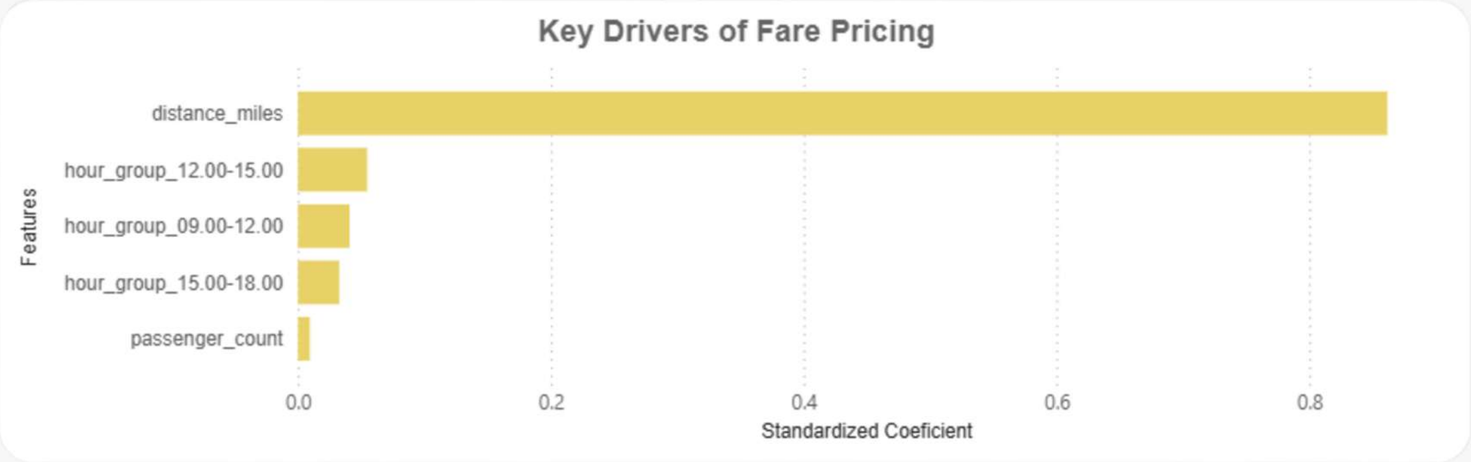
Visualization

Dashboard and presentation development

PRICING ANALYSIS

KEY DRIVERS OF FARE

Distance is the **most influential variable** of fare based on standardized regression coefficients.



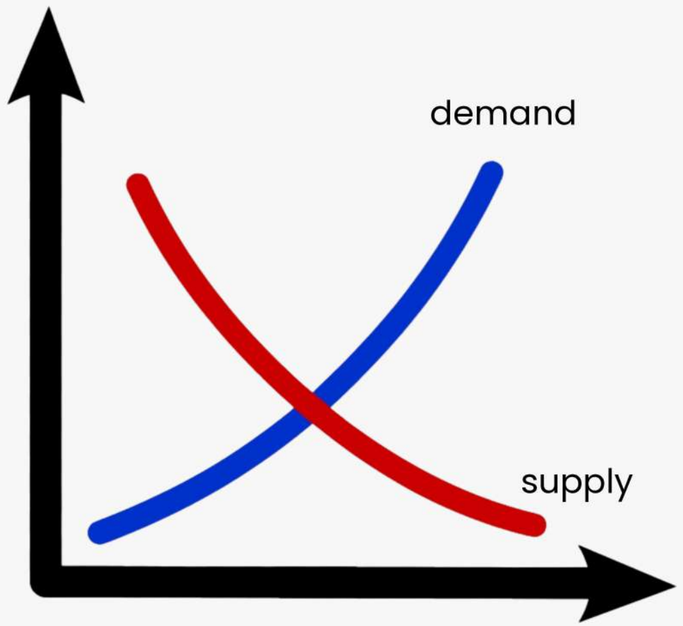
PRICING STRATEGY PRINCIPLE

DEMAND-BASED PRICING

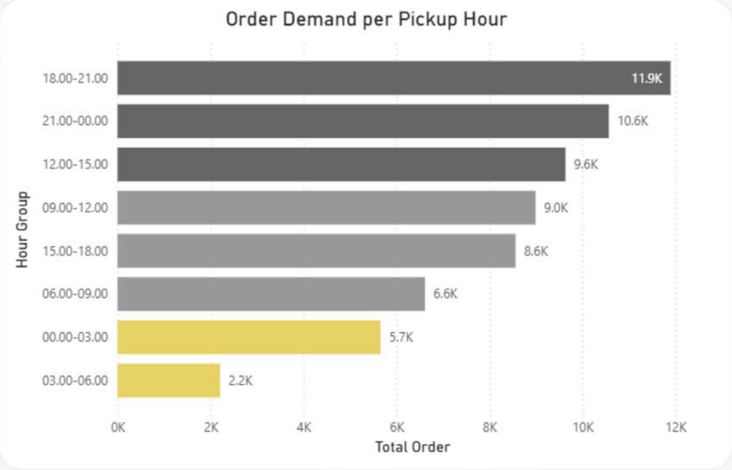
An effective pricing model aligns with **market demand dynamics.**

- High Demand → Higher Fares
- Low Demand → Discounted Fares

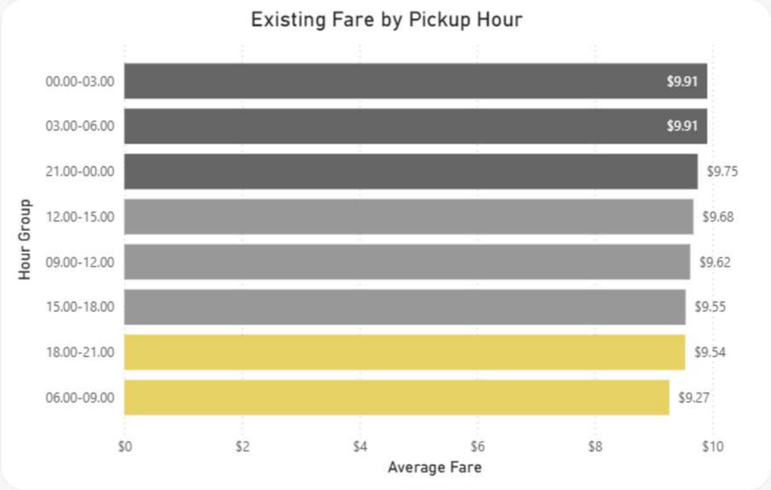
This strategy optimizes revenue generation while ensuring consistent demand levels and market competitiveness.



PICKUP HOUR ANALYSIS



Evaluation



Problem: Existing pricing fails to capture maximum value during high demand hours.

Solution:

Apply **Surge Pricing** on the **most demanded hours**:

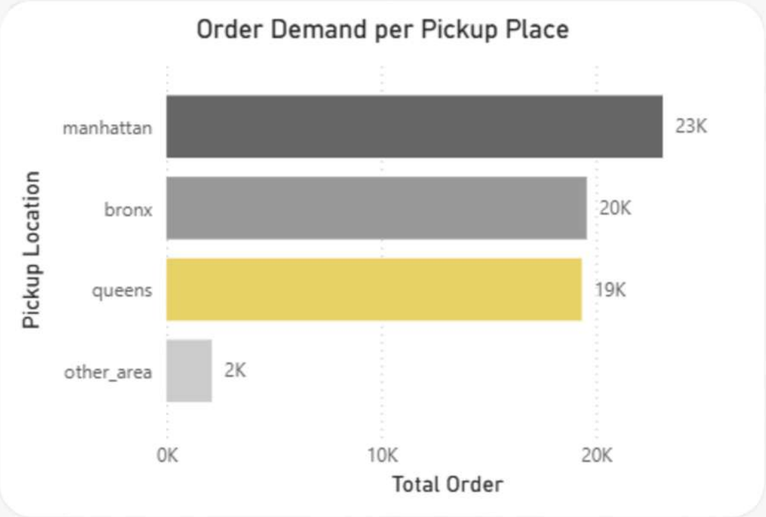
- 18:00-21:00
- 21:00-00:00
- 12:00-15:00

Apply **Discounts** on the **least demanded hours**:

- 00:00-03:00
- 03:00-06:00

Impact: Aligning price with demand will prevent revenue leakage and optimize booking yields.

PICKUP PLACE ANALYSIS



Problem: Existing pricing is misaligned with geography; "Other Area" is overpriced despite low demand.

Solution:

Apply High Fare:

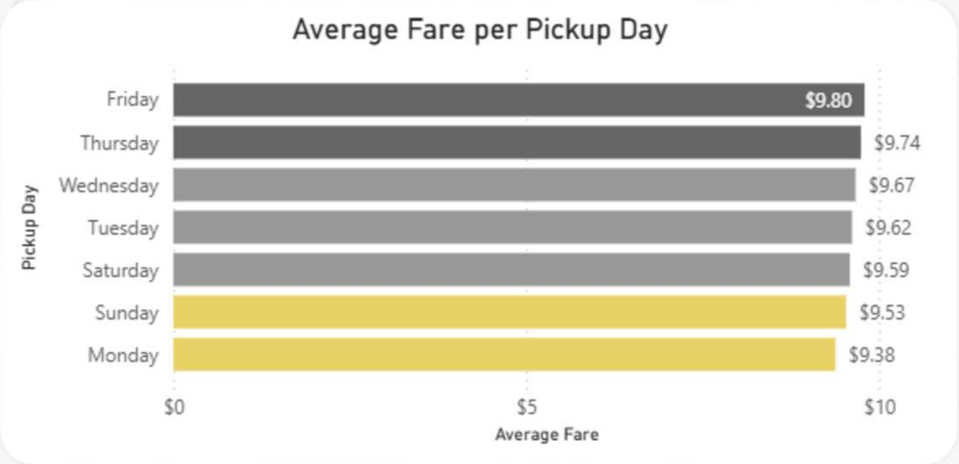
- Manhattan (Highest demand location).

Apply Discount Fare:

- Other Area (Lowest demand location).

Impact: Stimulates growth in underperforming zones while protecting margins.

PICKUP DAY ANALYSIS



Problem: High-demand days (Saturday) are underpriced, while existing discounts are not fully optimized for the lowest demand days.

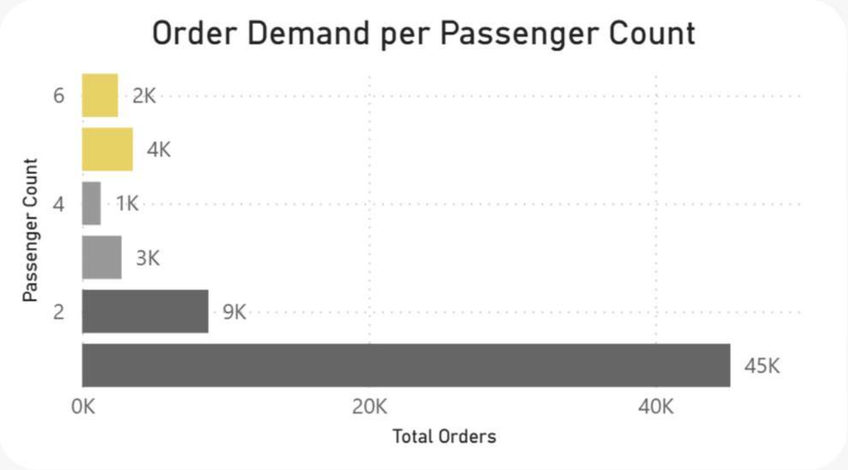
Solution:

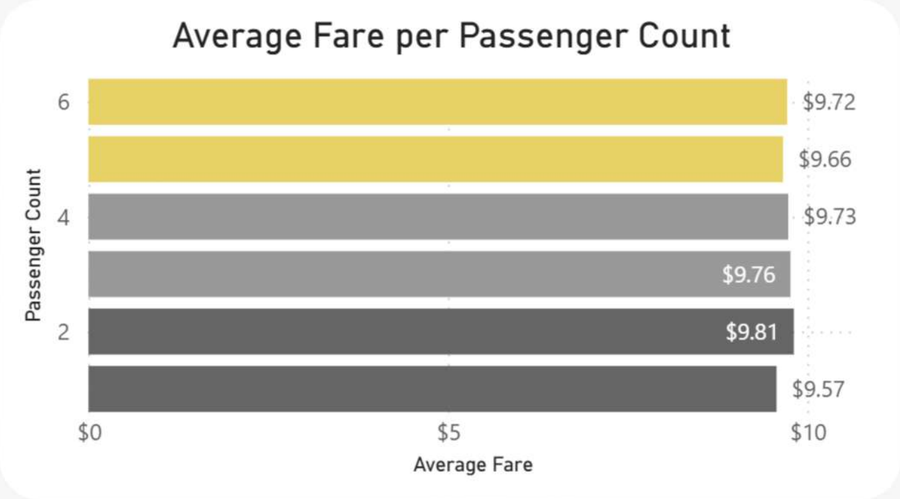
Apply High Fare:
- Saturday
- Friday
- Thursday

Apply Discount Fare:
- Sunday
- Monday.

Impact: Increasing Saturday rates to match demand will prevent revenue loss and stabilize daily earnings

PASSENGER COUNT ANALYSIS





Problem: Existing pricing is not optimized for vehicle size; orders for 5–6 passengers are charged the same rate as 1–4 passengers.

Solution: Apply Extra Fare: Orders with more than 4 passengers.
Reasoning: Large groups require bigger vehicles (higher operational cost).

Impact: Capacity-based surcharge ensures fair pricing and captures the added value of larger vehicle requirements.

IMPACT ANALYSIS

OPTIMIZED PRICING MODEL

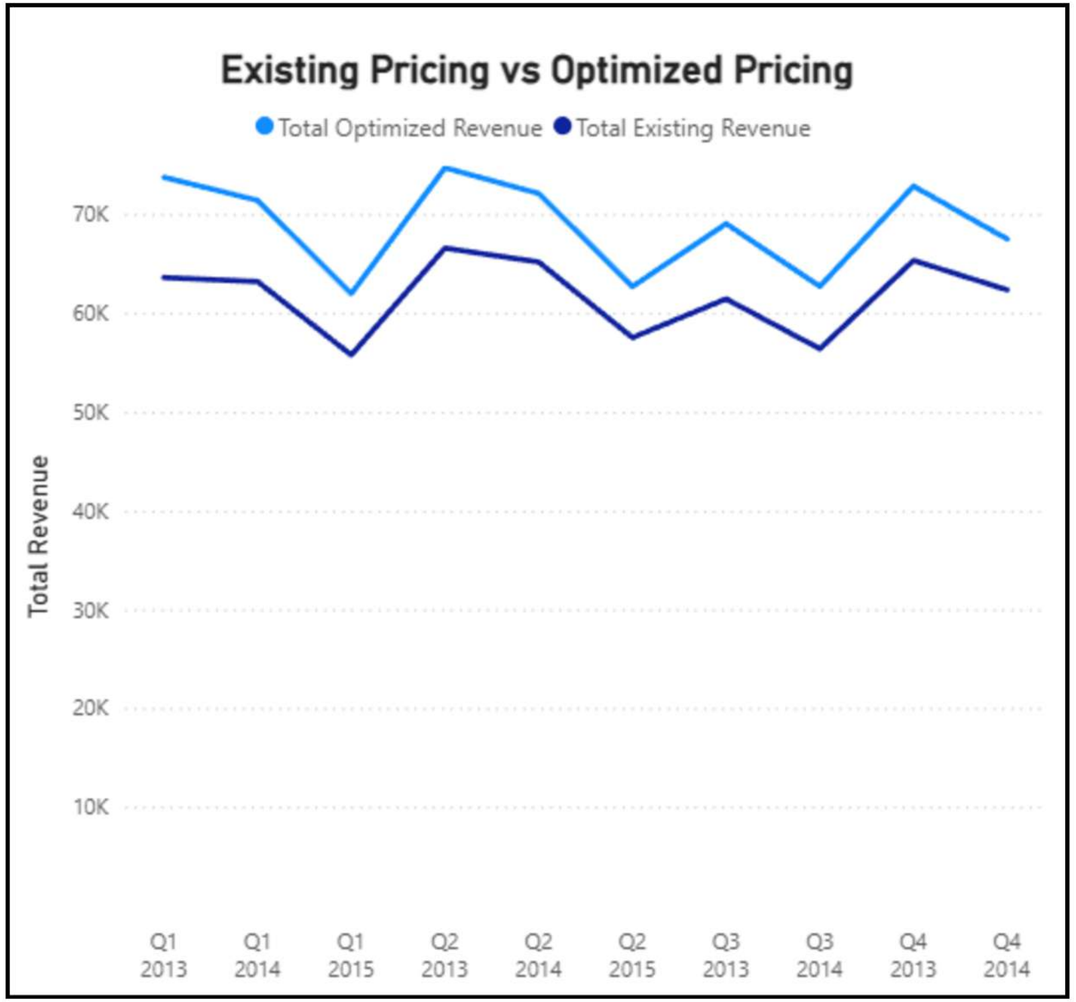
Variable	High Fare	Normal Fare	Discount Fare
Pickup Hour	18:00 – 00:00	06:00 – 18:00	00:00 – 06:00
Pickup Place	Manhattan	Bronx / Queens	Other Area
Pickup Day	Wed / Thu / Fri / Sat (PM)	Mon – Sun (Day)	Mon – Thu (AM)
Passenger Count	> 4	<= 4	—
Trip Distance	—	All Distances	—

This optimized model ensures that:

- High Fares are strategically applied to peak market conditions
- Discount Fares are utilized to stimulate growth in low-demand segments, ultimately maximizing total booking yield.

REVENUE PROJECTION (2013 – 2015)

Based on historical data, applying the optimized pricing model results in a **11.5% increase** in total revenue compared to the existing pricing structure.



Model Assumptions & Pricing Logic

This improvement is achieved under the following conditions:

- Same Base Pricing: Base fare and distance rates remain unchanged.
- Demand-Based Pricing: Prices are adjusted based on time, location, day, and passenger count.
- Peak Pricing: A 20% increase is applied during high-demand periods.
- Low-Demand Strategy: A 20% discount is applied during low-demand periods to encourage more bookings.

STRATEGIC INSIGHTS

Distance as Key Driver

Distance remains the most influential factor in pricing and should be prioritized in optimization strategies.

Revenue Maximization

Higher fares should be applied during peak demand periods to capture additional revenue.

Demand Stimulation

Discounted pricing during low-demand periods can help increase booking volume and reduce idle capacity.

Pricing Gaps Identified

The current pricing model is not fully aligned with demand patterns:

- Pickup Hour: Peak periods are not consistently priced higher
- Pickup Location: Low-demand areas lack proper incentives
- Pickup Day: No dynamic pricing between weekdays and weekends
- Passenger Count: No pricing adjustment for larger groups

BUSINESS RECOMMENDATIONS

1

MARKETING STRATEGY

Launch campaigns to increase Average Distance per Order through:

- Distance-based loyalty points and achievements.
- Tiered price reductions for long-haul trips (e.g., every 10 miles).

3

DRIVER PARTNERS

Provide demand-condition guidance to drivers, enabling them to position themselves in High Fare zones to maximize their own earnings and platform revenue.

2

PRODUCT & PRICING TEAM

Implement a ±20% Dynamic Pricing model (High/Discount) across all demand variables to achieve the projected 11.5% revenue growth.

4

PARTNER RELATIONS

Maintain driver supply during low-demand periods by offering targeted small incentives to prevent driver shortages and ensure consistent service availability.

LINKS

1. Dataset from Kaggle
2. Python Code (Google Colab)
3. Power BI Dashboard